

Parity Rock–Paper–Scissors Games and the Alper Constant

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5 October 2026

Abstract

Two players choose numbers from $\{1, \dots, n\}$. Equal numbers draw, an odd difference is won by the smaller number, a nonzero even difference is won by the larger number, and a win with the number m pays $2m - h$, where $0 \leq h < 2$ is fixed. For $h = 1$ this is the game of Problem 1012 of Project Euler; for $h = 0$ the payment is proportional to the winning number. For every h outside an explicit countable set we prove that the game has a unique equilibrium, supported on $2k + 1$ consecutive numbers ending at n or $n - 1$, and we solve it in closed form. When the support ends at n , the probability $P(n)$ of the largest number is $((2k^2 + 1)R_k - 2(k^2 - 1))/(3(2k + 1))$ for an explicit product R_k that depends only on the largest payment; otherwise $P(n) = 0$. The support widens at thresholds given by the zeros of these functions; for $h \in \{0, 1\}$ the thresholds are cubic polynomials in k for each parity. For $S_h(N) = \sum_{n=3}^N P(n)$ and $h \in \{0, 1\}$ we prove $S_h(N) = (3/2)^{4/3} N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) + O(N^{-1/3})$, where $\tau \in [0, 1)$ is the relative position of N in its block, and we determine the next term. The constant $\mathcal{A}(h)$ plays the role that Euler’s constant plays for the harmonic series. We call $\mathcal{A} = \mathcal{A}(0) = 1.7771799879\dots$ the Alper constant, give a series and an integral representation, and certify 80 decimal places of $\mathcal{A}$ and of $\mathcal{A}(1) = 1.6464176534\dots$ with interval arithmetic. No expression of either constant in classical constants was found.

1 Introduction

In rock–paper–scissors every option beats exactly one other option. With equal payments the unique equilibrium is uniform. Unequal payments change the picture in a counterintuitive way. Suppose the loser pays 5 to a winning rock, 3 to a winning paper and 1 to winning scissors. Then the unique equilibrium plays rock, paper and scissors with probabilities $\frac{1}{9}$, $\frac{5}{9}$ and $\frac{1}{3}$: the most lucrative option is used least.

We study the following family of extensions to n options. Two players choose numbers from $\{1, \dots, n\}$ simultaneously. Equal numbers draw. If the numbers differ by an odd amount, the smaller number wins. If they differ by a nonzero even amount, the larger number wins. A win with the number m is paid $2m - h$ by the loser, where $0 \leq h < 2$ is a fixed parameter. We write $G_n(h)$ for this game. Every increasing payment schedule $c(m - \theta)$ with $c > 0$ and $0 \leq \theta < 1$ is covered after scaling, with $h = 2\theta$. Two members stand out:

- $h = 1$: the payments are 1, 3, 5, $\dots$. For $n = 3$ the numbers 1, 2, 3 play the roles of scissors, paper and rock with the payments 1, 3, 5 above. This game was posed as Problem 1012 of Project Euler [8], and a solution is described in [2].

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- $h = 0$: the payment is proportional to the winning number. This is the simplest member, and the one whose constant we call the Alper constant.

Whenever the equilibrium is unique, let $P(n)$ be the probability that it assigns to the largest number n , and put

$$S_h(N) = \sum_{n=3}^N P(n).$$

In particular, these quantities are unambiguous for $h \notin \mathcal{E}$, where the exceptional set $\mathcal{E}$ is defined below.

Solving $G_n(h)$ numerically for moderate n and generic h suggests that the equilibrium plays an odd number of consecutive numbers and that its width grows by two at isolated values of n . Sampled ratios $S_0(N)/N^{1/3}$ approach their limit slowly: the ratio is 1.330 at $N = 10^3$ and 1.613 at $N = 10^5$, while its limit turns out to be $(3/2)^{4/3} \approx 1.7171$. We prove the support and transition laws for every $h \notin \mathcal{E}$, and the asymptotic formulas for $h \in \{0, 1\}$. The latter isolate the constant that governs the slow approach.

Throughout, $\log$ is the natural logarithm, γ is Euler's constant and $c_j = \binom{2j}{j} 4^{-j}$. For $k \geq 1$ and real y put

$$R_k(y) = \prod_{j=1}^k \frac{y - 2j}{y - 2j + 1}, \quad P_k(y) = \frac{(2k^2 + 1) R_k(y) - 2(k^2 - 1)}{3(2k + 1)}. \quad (1)$$

The variable y will be half of the largest payment in play, so $y = n - \frac{h}{2}$ when the largest number is n . For $k \geq 2$ the function P_k increases on $y > 2k$ and has exactly one zero β_k there (Lemma 5.2). Put $t_1(h) = 3$ and $t_k(h) = \lfloor \beta_k + \frac{h}{2} \rfloor + 1$ for $k \geq 2$: this is the first integer n with $P_k(n - \frac{h}{2}) > 0$. The integers n with $t_k(h) \leq n \leq t_{k+1}(h) - 1$ form the *block* k ; the blocks partition the integers $n \geq 3$. Finally let $\mathcal{E}$ be the set of $h \in [0, 2)$ for which $\beta_k + \frac{h}{2}$ is an integer for some $k \geq 2$. It is countable, and it contains neither 0 nor 1.

Theorem A (Theorems 4.3 and 5.4). *Let $0 \leq h < 2$, $h \notin \mathcal{E}$, and $n \geq 3$. The game $G_n(h)$ has a unique equilibrium. Let k be the block of n . If $n \leq t_{k+1}(h) - 2$, the support is $\{n - 2k, \dots, n\}$ and $P(n) = P_k(n - \frac{h}{2}) > 0$. If $n = t_{k+1}(h) - 1$, the support is $\{n - 1 - 2k, \dots, n - 1\}$ and $P(n) = 0$. The whole equilibrium is given in closed form by Theorem 4.3. For the two main cases*

$$t_k(0) = \left\lceil \frac{4k^3 + 5k}{6} \right\rceil + 1, \quad t_k(1) = \left\lfloor \frac{4k^3 + 5k}{6} \right\rfloor + 2.$$

The first values are $t_k(0) = 3, 8, 22, 47, 89, 150, 236, 349$ and $t_k(1) = 3, 9, 22, 48, 89, 151, 236, 350$.

Theorem B (Theorem 7.3). *Let $h \in \{0, 1\}$ and $\kappa = (3/2)^{4/3} = \sqrt[3]{81/16}$. For $N \geq 3$ let K be defined by $t_K(h) - 1 \leq N < t_{K+1}(h) - 1$, and put $\tau = (N - t_K(h) + 1)/(t_{K+1}(h) - t_K(h)) \in [0, 1)$. Then, uniformly in τ ,*

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O(N^{-2/3}).$$

Since $K = (3N/2)^{1/3} + O(1)$, the term with $1/K$ is of order $N^{-1/3}$. The quadratic term cannot be dropped: within every block, $S_h(N) - \kappa N^{1/3} + \frac{1}{4} \log N$ sweeps from about $-\mathcal{A}(h)$ down to about $-\mathcal{A}(h) - \frac{3}{8}$ and back. At the ends of the blocks, where P vanishes and $\tau = 0$, the oscillation disappears.

The Alper constant. The constant $\mathcal{A}(h)$ is an Euler-type constant. The harmonic numbers satisfy $H_N - \log N \rightarrow \gamma$; here, along the ends $N_K = t_{K+1}(h) - 1$ of the blocks,

$$\kappa N_K^{1/3} - \frac{1}{4} \log N_K - S_h(N_K) \rightarrow \mathcal{A}(h) \quad (K \rightarrow \infty).$$

We define the *Alper constant* as the constant of the simplest member, $h = 0$:

$$\mathcal{A} = \mathcal{A}(0) \approx 1.7771799879054231885280182162763415624255 \\ 7103207026437429288159344562194194456353$$

For the Project Euler game,

$$\mathcal{A}(1) \approx 1.6464176534405600338544841159858132863029 \\ 9683869991753007336025510008825020074857$$

Section 8 expresses both constants as rapidly computable series and as integral-series, and certifies that the values above are correctly rounded to 80 decimal places.

The case $h = 1$ is a Project Euler problem, and its solvers may have obtained parts of our results for it, including the closed form and the first-order asymptotic formula; we make no priority claim for this case. The contributions developed here include the family $G_n(h)$, its transition law for general h , proofs of the asymptotic formulas for $h \in \{0, 1\}$, the second-order term, the Alper constant and the certified digits. Section 2 sets up the games, Section 3 shows that the supports are intervals, and Section 4 solves the equilibrium equations on an interval of odd length. Section 5 proves the transition law and Theorem A. Section 6 sums the probabilities over blocks. Section 7 proves Theorem B, Section 8 studies the constants, and Section 9 compares the theorems with exact values of $S_0(10^m)$ for $m \leq 20$.

2 The games

Let a_{ij} be the payment received by a player who chooses i when the opponent chooses j . It is 0 for $i = j$. Otherwise the winning number w is $\min(i, j)$ when $i - j$ is odd and $\max(i, j)$ when $i - j$ is even, and the winner receives $2w - h$.

Lemma 2.1. *For all i, j , $a_{ij} = (i - j) + \operatorname{sgn}(i - j)(-1)^{i+j}(i + j - h)$. In particular $a_{ji} = -a_{ij}$.*

Proof. Take $j < i$. If $i - j$ is even, i wins and the formula gives $(i - j) + (i + j - h) = 2i - h$. If $i - j$ is odd, j wins and it gives $(i - j) - (i + j - h) = -(2j - h)$. The case $j > i$ follows by antisymmetry. $\square$

Against a vector $p = (p_1, \dots, p_n)$ the pure option i earns $f_i = \sum_j a_{ij}p_j$, and we use this formula for every integer i . Since $A = (a_{ij})$ is skew-symmetric, $p^\top A p = 0$, and the value of the game is 0. Hence a probability vector p is an equilibrium exactly when

$$p_i \geq 0, \quad \sum_i p_i = 1, \quad f_i \leq 0 \text{ for all } i, \quad f_i = 0 \text{ whenever } p_i > 0; \quad (2)$$

the last condition follows from the others because $\sum_i p_i f_i = p^\top A p = 0$. An equilibrium exists for every n by the minimax theorem [9]. The payoff rule does not depend on n , so the matrix of $G_{n-1}(h)$ is a principal submatrix of the matrix of $G_n(h)$.

For $n = 3$ the payoffs are $f_1 = (2 - h)p_2 - (6 - h)p_3$, $f_2 = -(2 - h)p_1 + (4 - h)p_3$ and $f_3 = (6 - h)p_1 - (4 - h)p_2$. Setting them to zero and normalising gives

$$p = \frac{(4 - h, 6 - h, 2 - h)}{12 - 3h},$$

so $p = (\frac{1}{3}, \frac{1}{2}, \frac{1}{6})$ for $h = 0$ and $p = (\frac{1}{3}, \frac{5}{9}, \frac{1}{9})$ for $h = 1$, the weighted rock–paper–scissors equilibrium of the introduction.

3 Interval supports

For a probability vector p let X_i be the probability that option i wins: the total mass of the numbers $j < i$ of the parity of i and of the numbers $j > i$ of the other parity.

Lemma 3.1. *For every probability vector p and every $i \leq n - 2$,*

$$f_{i+2} - f_i = 4(X_i - p_{i+1}) + (2i + 4 - h)(p_i + p_{i+2}) - (4i + 2 - 2h)p_{i+1}, \quad X_i \geq p_{i+1}.$$

Proof. The options i and $i + 2$ have the same parity, so they obtain the same result against every $j \notin \{i, i+1, i+2\}$. When both win, $i+2$ receives 4 more. The three remaining columns give $a_{i+2,i} - a_{ii} = 2i + 4 - h$, $a_{i+2,i+1} - a_{i,i+1} = -(2i + 2 - h) - (2i - h) = -(4i + 2 - 2h)$ and $a_{i+2,i+2} - a_{i,i+2} = 2i + 4 - h$. Option i beats $i + 1$, which gives $X_i \geq p_{i+1}$. $\square$

Proposition 3.2. *Let $n \geq 3$ and let p be an equilibrium of $G_n(h)$. Its support is an interval $\{a, \dots, b\}$ with $b \in \{n - 1, n\}$ and $b > a$.*

Proof. Suppose $p_i > 0$ and $p_{i+1} = 0$ for some $i \leq n - 2$. Then $f_i = 0$ and Lemma 3.1 gives $f_{i+2} \geq (2i + 4 - h)p_i > 0$, which contradicts (2). So every chosen number below $n - 1$ is followed by another chosen number, and the support is an interval $\{a, \dots, b\}$ with $b \geq n - 1 \geq 2$. A single number $b \geq 2$ is not an equilibrium, because $b - 1$ beats it and receives $2b - 2 - h > 0$. $\square$

For $h \notin \mathcal{E}$ the support has odd size, as proved in Proposition 5.3. For the payments $1, 3, 5, \dots$ of $h = 1$ there is a short argument modulo 2, but for general nonexceptional h the parity of the support follows from the explicit solution of the next section.

4 The equilibrium on an interval of odd length

Fix $k \geq 1$, $b \geq 2k + 1$, $a = b - 2k$, $L = 2k + 1$, $I = \{a, \dots, b\}$ and $y = b - \frac{h}{2}$, so that $2y$ is the payment for a win with b . For a real vector p supported on I put

$$s = \sum_j p_j, \quad M = \sum_j j p_j, \quad q_i = \sum_{j \leq i} (-1)^{i-j} p_j \quad (i \in \mathbb{Z}), \quad Q = q_b.$$

Thus $q_i = 0$ for $i < a$, $q_i = (-1)^{i-b} Q$ for $i \geq b$, and $p_i = q_i + q_{i-1}$. The numbers of the parity of b carry the mass $\frac{s+Q}{2}$, and the others carry $\frac{s-Q}{2}$.

Lemma 4.1 (moment equation). *For $a \leq j \leq b$,*

$$\sum_{i=a}^b a_{ij} = L \left(\frac{a+b}{2} - j \right) + \begin{cases} k, & j \equiv b \pmod{2}, \\ -(k+1), & j \not\equiv b \pmod{2}, \end{cases}$$

independently of h . Consequently, if $f_i = 0$ for all $i \in I$, then $L(\frac{a+b}{2}s - M) + k\frac{s+Q}{2} - (k+1)\frac{s-Q}{2} = 0$; for $s = 1$ this reads $M - \frac{Q}{2} = \frac{a+b}{2} - \frac{1}{2L}$, and for $s = 0$ it reads $M = \frac{Q}{2}$.

Proof. The column sum follows from Lemma 2.1 by pairing consecutive terms; the terms containing h cancel. Adding the L equations $f_i = 0$ gives the stated equation. $\square$

Lemma 4.2 (neighbour identity). *For every real vector p supported on I and every integer i ,*

$$f_i + f_{i+1} = (2i + 1)s - 2M + (-1)^{b-i} Q + (2i + 2 - h)q_{i+1} - (2i - h)q_{i-1}.$$

Proof. With $\varepsilon_j = (-1)^j$, Lemma 2.1 gives $f_i = is - M + \varepsilon_i[(i-h)T_i + U_i]$, where $T_i = \sum_{j < i} \varepsilon_j p_j - \sum_{j > i} \varepsilon_j p_j$ and $U_i = \sum_{j < i} \varepsilon_j j p_j - \sum_{j > i} \varepsilon_j j p_j$. Passing from i to $i+1$ changes T_i and U_i only in the terms of p_i and p_{i+1} , and $\varepsilon_{i+1} = -\varepsilon_i$, so $f_i + f_{i+1} = (2i+1)s - 2M - \varepsilon_i T_i - (2i+1-h)p_i + (2i+2-h)p_{i+1}$. Finally $\varepsilon_i T_i = q_i - q_{i-1} - (-1)^{b-i}Q$, and $p_i = q_i + q_{i-1}$, $p_{i+1} = q_{i+1} + q_i$. $\square$

If $s = 1$ and $f_i = 0$ on I , the identity for $a \leq i \leq b-1$ gives the $2k$ neighbour equations

$$(2i+2-h)q_{i+1} - (2i-h)q_{i-1} = 2M - (2i+1) - (-1)^{b-i}Q \quad (a \leq i \leq b-1). \quad (3)$$

Each equation links q_{i-1} and q_{i+1} . The unknowns therefore split into the chain $a-1, a+1, \dots, b-1$ and the chain $a, a+2, \dots, b$.

Theorem 4.3. *Let p be a real vector supported on I with $\sum_j p_j = 1$ and $f_i = 0$ for all $i \in I$. Put $R = R_k(y)$, $c_h = \frac{1-h}{3}$ and*

$$K = \frac{a+b-1}{2} - \frac{1}{2L}, \quad Q = \frac{2y(R-1) + 2k}{3} + \frac{1}{L}.$$

Then $p_m = q_m + q_{m-1}$ with

$$q_m = K + c_h - \frac{m}{3} - \left(K + c_h - \frac{a-1}{3}\right) \prod_{\substack{a \leq i \leq m-1 \\ i \equiv b \pmod{2}}} \frac{2i-h}{2i+2-h} \quad (m = a-1, a+1, \dots, b-1), \quad (4)$$

$$q_m = K + c_h + Q - \frac{m}{3} + \left(\frac{b}{3} - K - c_h\right) \prod_{\substack{m+1 \leq i \leq b-1 \\ i \not\equiv b \pmod{2}}} \frac{2i+2-h}{2i-h} \quad (m = a, a+2, \dots, b). \quad (5)$$

In particular

$$p_b = P_k(y), \quad p_a = K + c_h + Q - \frac{a}{3} + \left(\frac{b}{3} - K - c_h\right) \frac{yR}{y-2k}.$$

Proof. On the first chain the equations (3) have $(-1)^{b-i} = 1$. The homogeneous step multiplies by $\frac{2i-h}{2i+2-h}$. A function $\gamma' - \frac{m}{3}$ is a particular solution exactly when $2\gamma' = 2M - 1 - Q + \frac{2-2h}{3}$, because $(2i+2-h)(i+1) - (2i-h)(i-1) = 6i+2-2h$. This gives $\gamma' = M - \frac{1+Q}{2} + c_h = K + c_h$, by Lemma 4.1. Starting from $q_{a-1} = 0$ gives (4). Over the whole chain the product is $R_k(y)$.

On the second chain $(-1)^{b-i} = -1$, the particular solution is $K + c_h + Q - \frac{m}{3}$, and the chain ends at $q_b = Q$. Running it downwards gives (5). The full product over the second chain is $1/R_k(y+1)$, and the telescoping identity

$$R_k(y) R_k(y+1) = \frac{y-2k}{y} \quad (6)$$

gives the formula for $p_a = q_a$.

It remains to determine Q . Add the equations (3) for $a \leq i \leq b-1$:

- the left sides telescope to $(2b-h)Q + (2b-2-h)q_{b-1} - (2a+2-h)q_a - 2\sum_{m=a+1}^{b-2} q_m$;
- the terms $\pm Q$ on the right cancel, and $\sum_{i=a}^{b-1} (2i+1) = b^2 - a^2$.

The normalisation $1 = Q + 2\sum_{m=a}^{b-1} q_m$ removes the remaining sum. Together with Lemma 4.1 this yields

$$2y p_b - (2a-h)p_a = (2k-1)Q + \frac{1}{L}, \quad p_b = Q + q_{b-1}. \quad (7)$$

Inserting the two chain solutions leaves a linear equation for Q in which Q has the coefficient L . Its solution is the stated value. Then $p_b = Q + q_{b-1} = P_k(y)$. $\square$

Lemma 4.4 (the kernel). *For every $0 \leq h < 2$ the kernel of the principal submatrix A_I is a line, spanned by a vector with nonzero coordinate sum. Consequently the formulas of Theorem 4.3 define a vector with $\sum_j p_j = 1$ and $f_i = 0$ on I , and it is the only such vector.*

Proof. A skew-symmetric matrix of odd order is singular, so A_I has a nonzero kernel vector w . Suppose that $\sum_j w_j = 0$. Lemma 4.1 gives $M = \frac{Q}{2}$, and Lemma 4.2 turns the equations $f_i + f_{i+1} = 0$, $a \leq i \leq b-1$, into $(2i+2-h)q_{i+1} - (2i-h)q_{i-1} = 2M - (-1)^{b-i}Q$. On the first chain the right side is 0, so $q \equiv 0$ there, because $q_{a-1} = 0$. On the second chain the right side is $2Q$; the constant Q is a solution, and since $q_b = Q$ the solution is this constant. Hence $w_m = q_m + q_{m-1} = Q$ for all $m \in I$, so $0 = \sum_j w_j = LQ$ and $w = 0$, a contradiction. Thus every nonzero kernel vector has nonzero sum and can be normalised; Theorem 4.3 shows that the normalised kernel vector is unique, so the kernel is a line. $\square$

Example 4.5. Take $h = 0$, $b = 10$ and $k = 2$, so $I = \{6, \dots, 10\}$, $y = 10$ and $R_2(10) = \frac{8}{9} \cdot \frac{6}{7} = \frac{16}{21}$. The theorem gives

$$(p_6, p_7, p_8, p_9, p_{10}) = \left(\frac{29}{315}, \frac{92}{315}, \frac{34}{105}, \frac{74}{315}, \frac{2}{35} \right),$$

with $Q = -\frac{17}{315}$ and $M = \frac{496}{63}$, so that $M - \frac{Q}{2} = \frac{79}{10} = \frac{6+10}{2} - \frac{1}{10}$. This is the equilibrium of $G_{10}(0)$, and $P(10) = \frac{2}{35}$.

5 Neighbouring options and the transition law

The interval solution decides its own fate through the two options next to the support.

Proposition 5.1. *In the setting of Theorem 4.3,*

$$f_{a-1} = \frac{2k+3}{2k+1} 2y R_k(y) P_{k+1}(y+1), \quad f_{b+1} = -\frac{2k+3}{2k+1} (2y+2) P_{k+1}(y+2).$$

Proof. Lemma 4.2 with $i = a-1$ (where $f_a = 0$, $q_{a-2} = 0$ and $(-1)^{b-a+1} = -1$) and with $i = b$ (where $f_b = 0$, $q_{b+1} = -Q$ and $q_{b-1} = p_b - Q$) gives

$$f_{a-1} = (2a-1) - 2M - Q + (2a-h)p_a, \quad f_{b+1} = (2b+1) - 2M - Q - 2y p_b.$$

By Lemma 4.1, $2M + Q = a + b - \frac{1}{L} + 2Q$, and by (7),

$$f_{a-1} = 2y p_b - L(1+Q), \quad f_{b+1} = L + \frac{1}{L} - 2Q - 2y p_b.$$

Inserting $p_b = P_k(y)$ and the value of Q from Theorem 4.3, both payoffs become affine in $R = R_k(y)$:

$$\begin{aligned} 3L f_{a-1} &= (2k^2 + 4k + 3)(2y - 4k - 2) - 2k(k+2) 2y R, \\ 3L f_{b+1} &= 2k(k+2)(2y+2) - (2k^2 + 4k + 3) 2y R. \end{aligned}$$

On the other hand, the definition of R_{k+1} gives

$$R_{k+1}(y+1) R_k(y) = \frac{y-2k-1}{y}, \quad R_{k+1}(y+2) = \frac{y}{y+1} R_k(y).$$

Since $2(k+1)^2 + 1 = 2k^2 + 4k + 3$ and $2((k+1)^2 - 1) = 2k(k+2)$, the definition (1) of P_{k+1} turns the right-hand sides into $3(2k+3) 2y R_{k+1}(y+1)$ and $-3(2k+3)(2y+2)P_{k+1}(y+2)$. $\square$

Lemma 5.2 (root bound). *Let $k \geq 2$. On the half-line $y > 2k$ the function P_k is continuous and strictly increasing, and its unique zero β_k satisfies*

$$\frac{4k^3 + 5k + 3}{6} < \beta_k < \frac{4k^3 + 5k + 6}{6}.$$

In particular $\beta_k \notin \frac{1}{2}\mathbb{Z}$. For $k = 1$ we have $P_1(y) = \frac{y-2}{3(y-1)} > 0$ for $y > 2$.

Proof. For $y > 2k$ every factor of $R_k(y)$ is positive and increasing, so R_k increases strictly from 0 to 1, and $P_k(y) = 0$ exactly when $-\log R_k(y) = \lambda_k := \log \frac{2k^2+1}{2k^2-2}$. Put $x = y - k - \frac{1}{2}$ and

$$U = \bigcup_{j=1}^k [k - 2j + \frac{1}{2}, k - 2j + \frac{3}{2}], \quad G(x) = \int_U \frac{dt}{x+t}.$$

The interval with index j contributes $\log \frac{y-2j+1}{y-2j}$, so $G(x) = -\log R_k(y)$. The set U is symmetric about 0, has measure k and lies in $[-d, d]$ with $d = k - \frac{1}{2}$; on $x > d$ the function G is strictly decreasing. Let x^* be the solution of $G(x^*) = \lambda_k$, so that $\beta_k = x^* + k + \frac{1}{2}$.

Lower bound. Let $x_0 = k(4k^2 - 1)/6 > d$. Then $x_0 + \frac{k}{2} = \frac{k(2k^2+1)}{3}$ and $x_0 - \frac{k}{2} = \frac{k(2k^2-2)}{3}$, so $\int_{-k/2}^{k/2} \frac{dt}{x_0+t} = \lambda_k$. Folding both integrals onto $t \geq 0$ gives

$$G(x_0) - \lambda_k = \int_{U_+} g(t) dt - \int_0^{k/2} g(t) dt, \quad g(t) = \frac{2x_0}{x_0^2 - t^2}, \quad U_+ = U \cap [0, \infty).$$

The set U_+ has measure $\frac{k}{2}$, and since $k \geq 2$ it has positive measure beyond $\frac{k}{2}$. As g is strictly increasing on $[0, x_0)$, the difference is positive. Hence $G(x_0) > \lambda_k$ and $x^* > x_0$.

Upper bound. Let $z = x_0 + \frac{1}{2}$. By symmetry $G(z) = \int_U \frac{z dt}{z^2 - t^2}$, and for $|t| \leq d$ we have $\frac{z}{z^2 - t^2} \leq \frac{1}{z} (1 + \frac{t^2}{z^2 - d^2})$. Since $\int_U t^2 dt = kv$ with $v = \frac{4k^2-3}{12}$,

$$G(z) \leq \frac{k}{z} \left(1 + \frac{v}{z^2 - d^2} \right) < \frac{k}{x_0}.$$

The second inequality is equivalent to $2x_0v < z^2 - d^2 = x_0^2 + x_0 - (k^2 - k)$, which follows from $x_0 > 2v$ and $x_0 > k^2 - k$. Finally $\frac{k}{x_0} = \frac{6}{4k^2-1} = \frac{2u}{2+u}$ with $u = \frac{3}{2k^2-2}$, and $\frac{2u}{2+u} < \log(1+u) = \lambda_k$. Hence $G(z) < \lambda_k$ and $x^* < z$.

Thus $x_0 < x^* < x_0 + \frac{1}{2}$, which is the stated bound because $x_0 + k + \frac{1}{2} = \frac{4k^3+5k+3}{6}$. The number $\frac{4k^3+5k}{6}$ is an integer for even k and a half-integer for odd k , so the two bounds are consecutive points of $\frac{1}{2}\mathbb{Z}$. $\square$

By Lemma 5.2, $t_k(h) = \lfloor \beta_k + \frac{h}{2} \rfloor + 1$ is the first integer n with $P_k(n - \frac{h}{2}) > 0$, and for $h \notin \mathcal{E}$ no integer n gives $P_k(n - \frac{h}{2}) = 0$. Since $\beta_k \notin \frac{1}{2}\mathbb{Z}$, neither 0 nor 1 lies in $\mathcal{E}$, and the bounds of Lemma 5.2 give the formulas for $t_k(0)$ and $t_k(1)$ stated in Theorem A, equivalently

$$t_k(h) = \frac{2}{3}k^3 + \frac{5}{6}k + \frac{5 + 2h - (-1)^{k+h}}{4} \quad (h \in \{0, 1\}). \quad (8)$$

Proposition 5.3. *Let $h \notin \mathcal{E}$. Every equilibrium of $G_n(h)$ has a support of odd size $2k + 1 \geq 3$.*

Proof. Suppose the support $I_0 = \{a, \dots, b\}$ of an equilibrium p has even size $2k \geq 2$, and let $I = \{a, \dots, b+1\}$; the option $b+1$ may lie outside the game, and f_{b+1} is computed with Lemma 2.1. On I we have $A_I p = \varphi e_{b+1}$ with $\varphi = f_{b+1}$, because $f_i = 0$ on I_0 . Let v be the normalised kernel vector of A_I from Lemma 4.4. By Theorem 4.3, $v_{b+1} = P_k(b+1 - \frac{h}{2})$, which is nonzero: for $k \geq 2$ because $h \notin \mathcal{E}$, and for $k = 1$ by Lemma 5.2. Skew-symmetry gives $0 = -(A_I v)^\top p = v^\top A_I p = \varphi v_{b+1}$, so $\varphi = 0$. Then p is a normalised kernel vector of A_I , so $p = v$ and $p_{b+1} = v_{b+1} \neq 0$, a contradiction. $\square$

Theorem 5.4. *Let $h \notin \mathcal{E}$ and let $n \geq 3$ lie in the block k . The game $G_n(h)$ has exactly one equilibrium. If $n \leq t_{k+1}(h) - 2$, its support is $\{n - 2k, \dots, n\}$ and $P(n) = P_k(n - \frac{h}{2}) > 0$. If $n = t_{k+1}(h) - 1$, its support is $\{n - 1 - 2k, \dots, n - 1\}$ and $P(n) = 0$.*

Proof. Write t_k for $t_k(h)$. Let p be an equilibrium of $G_n(h)$; it exists by the minimax theorem. By Propositions 3.2 and 5.3 its support is $\{a, \dots, b\}$ with $b \in \{n - 1, n\}$ and $b - a = 2k'$ for some $k' \geq 1$, and by Theorem 4.3, $p_b = P_{k'}(b - \frac{h}{2})$.

First, $t_{k'} \leq b \leq t_{k'+1} - 2$. Indeed $P_{k'}(b - \frac{h}{2}) = p_b > 0$, so $b \geq t_{k'}$. If $a \geq 2$, the option $a - 1$ exists and $f_{a-1} \leq 0$. As $R_{k'}(b - \frac{h}{2}) > 0$, Proposition 5.1 gives $P_{k'+1}(b + 1 - \frac{h}{2}) \leq 0$, hence < 0 , and so $b + 1 < t_{k'+1}$. If $a = 1$, then $b = 2k' + 1 \leq t_{k'+1} - 2$, because $t_{k'+1} > \beta_{k'+1} > 2k' + 3$ by Lemma 5.2.

If $b = n - 1$, the option $n = b + 1$ exists and $f_n \leq 0$, so $P_{k'+1}(b + 2 - \frac{h}{2}) \geq 0$ by Proposition 5.1, hence > 0 , and $b + 2 \geq t_{k'+1}$. Together with the first step, $n = t_{k'+1} - 1$; this is the last element of the block k' , so $k' = k$, $n = t_{k+1} - 1$ and $P(n) = p_n = 0$.

If $b = n$, the first step gives $t_{k'} \leq n \leq t_{k'+1} - 2$, so $k' = k$, $n \leq t_{k+1} - 2$ and $P(n) = p_n = P_k(n - \frac{h}{2}) > 0$.

The two cases are distinguished by whether $n = t_{k+1} - 1$, so the support is determined by n . By Lemma 4.4, p is the only normalised vector on this support with vanishing payoffs on it. Hence the equilibrium is unique. $\square$

Remark 5.5. The proof never checks that the closed-form probabilities are positive or that the excluded options lose. Positivity and the sign conditions are inherited from the equilibrium whose existence the minimax theorem guarantees; as a by-product, every p_m in Theorem 4.3 is positive when $t_k(h) \leq b \leq t_{k+1}(h) - 2$.

Write $\beta_k = \frac{4k^3 + 5k + 3}{6} + \vartheta_k$ with $0 < \vartheta_k < \frac{1}{2}$. Unwinding the definition of $t_k(h)$,

$$\mathcal{E} = \{1 - 2\vartheta_k : k \geq 2 \text{ even}\} \cup \{2 - 2\vartheta_k : k \geq 3 \text{ odd}\}.$$

For even k , $t_k(h)$ equals $t_k(0)$ when $h < 1 - 2\vartheta_k$ and $t_k(1)$ when $h > 1 - 2\vartheta_k$. For odd k , $t_k(0) = t_k(1)$, and $t_k(h)$ equals this common value when $h < 2 - 2\vartheta_k$ and exceeds it by one when $h > 2 - 2\vartheta_k$. In fact,

$$\vartheta_k = \frac{3}{8k} + O(k^{-3}).$$

To prove this, retain U , G and $x_0 = k(4k^2 - 1)/6$ from the proof of Lemma 5.2, and put $V = [-k/2, k/2]$. Then $\lambda_k = \int_V (x_0 + t)^{-1} dt$. Both sets are symmetric, have measure k and lie in $[-k, k]$. Expanding the integrands, with the odd moments cancelling, gives

$$G(x_0) - \lambda_k = \frac{\int_U t^2 dt - \int_V t^2 dt}{x_0^3} + O\left(\frac{k^5}{x_0^5}\right) = \frac{k(k^2 - 1)}{4x_0^3} + O\left(\frac{k^5}{x_0^5}\right).$$

Here $\int_U t^2 dt = k(4k^2 - 3)/12$ and $\int_V t^2 dt = k^3/12$. Uniformly for $x_0 \leq x \leq x_0 + 1/2$, symmetry also gives $-G'(x) = kx_0^{-2}(1 + O(k^{-3}))$. Since $G(x_0 + \vartheta_k) = \lambda_k$ and $0 < \vartheta_k < 1/2$, the mean value theorem yields

$$\vartheta_k = \frac{k^2 - 1}{4x_0} (1 + O(k^{-3})) + O\left(\frac{k^4}{x_0^3}\right) = \frac{3}{8k} + O(k^{-3}).$$

Consequently $\mathcal{E}$ accumulates only at 1 and 2, and for every fixed $h \notin \mathcal{E}$ the thresholds eventually agree with those of $h = 0$ (if $h < 1$) or of $h = 1$ (if $h \geq 1$). For illustration, $\vartheta_2 \approx 0.1458$, $\vartheta_{10} \approx 0.0372$ and $\vartheta_{400} \approx 0.00094$; monotonicity is not needed.

For $h \in \mathcal{E}$ uniqueness fails for at least one game: if $\beta_k + \frac{h}{2} = m \in \mathbb{Z}$, let nonexceptional parameters tend to h from both sides in Theorem 5.4. The limiting vectors are supported on $\{m - 2k + 1, \dots, m - 1\}$

and $\{m - 2k, \dots, m - 2\}$. They are equilibria of $G_{m-1}(h)$ because the equilibrium inequalities are continuous. They are distinct: the first assigns probability $P_{k-1}(\beta_k - 1) > 0$ to $m - 1$, while the second assigns zero. The strict inequality follows from Lemma 5.2, which gives $\beta_k - 1 > \beta_{k-1}$ for $k \geq 3$; for $k = 2$ use $P_1(y) > 0$ for $y > 2$. Their convex combinations are also equilibria.

6 Block sums

From now on $h \in \{0, 1\}$. By Theorem 5.4, S_h is a sum over blocks. For $k \geq 1$ put

$$\ell_k = t_{k+1}(h) - t_k(h) - 1 = 2k^2 + 2k + \frac{1 + (-1)^{k+h}}{2}, \quad B_k = B_k(h) = \sum_{n=t_k(h)}^{t_{k+1}(h)-2} P_k\left(n - \frac{h}{2}\right),$$

so that $S_h(t_{K+1}(h) - 1) = \sum_{k=1}^K B_k$. Table 1 lists the first blocks for $h = 0$.

k	$t_k(0)$	n with $P(n) = P_k(n) > 0$	zero of P	ℓ_k	$B_k(0)$
1	3	3–6	7	4	0.9055555556
2	8	8–20	21	13	1.2051685518
3	22	22–45	46	24	1.2449235895
4	47	47–87	88	41	1.3328846236
5	89	89–148	149	60	1.3441658999
6	150	150–234	235	85	1.3842642085

Table 1: The first blocks of the game $G_n(0)$. In the block k the support has $2k + 1$ elements; at the zero of P it is shifted down by one.

Lemma 6.1 (partial fractions). *For $k \geq 1$,*

$$1 - R_k(y) = \sum_{j=1}^k \frac{w_{k,j}/2}{y - 2j + 1}, \quad w_{k,j} = 2(2k - 2j + 1) c_{j-1} c_{k-j}, \quad \sum_{j=1}^k w_{k,j} = 2k.$$

Moreover $w_{k,j}/(2k) = \Pr(X = j - 1)$, where X has the beta-binomial distribution with parameters $k - 1$, $\frac{1}{2}$ and $\frac{3}{2}$, so that

$$P_k(y) = \frac{1}{2k+1} - \frac{2k(2k^2+1)}{3(2k+1)} \mathbb{E}\left[\frac{1}{2y-2-4X}\right]. \quad (9)$$

Proof. R_k is a rational function of y with simple poles at $y = 2j - 1$ and $R_k(\infty) = 1$. The factor with index j equals $1 - \frac{1}{y-2j+1}$. At its pole the other factors take the value $\prod_{i < j} \frac{2(j-i)-1}{2(j-i)} \prod_{i > j} \frac{2(i-j)+1}{2(i-j)} = c_{j-1}(2k - 2j + 1)c_{k-j}$, which gives the residues. Since $\sum_q (2q+1)c_q z^q = (1-z)^{-3/2}$ and $\sum_q c_q z^q = (1-z)^{-1/2}$, comparing coefficients in $(1-z)^{-3/2}(1-z)^{-1/2} = (1-z)^{-2}$ gives $\sum_{q=0}^{k-1} (2q+1)c_q c_{k-1-q} = k$. The beta-binomial probability of x is $\binom{k-1}{x} (\frac{1}{2})_x (\frac{3}{2})_{k-1-x} / (2)_{k-1}$ in Pochhammer notation [7]; with $(\frac{1}{2})_x = x! c_x$ and $(\frac{3}{2})_z = (2z+1) z! c_z$ this equals $c_x(2k - 2x - 1)c_{k-1-x}/k$. Finally $2(y - 2j + 1) = 2y - 2 - 4(j - 1)$ and $P_k = \frac{1}{2k+1} - \frac{2k^2+1}{3(2k+1)}(1 - R_k)$. $\square$

For example $w_{2,\cdot} = (3, 1)$, $w_{3,\cdot} = (\frac{15}{4}, \frac{3}{2}, \frac{3}{4})$ and $w_{4,\cdot} = (\frac{35}{8}, \frac{15}{8}, \frac{9}{8}, \frac{5}{8})$. Let ψ be the digamma function, $H_m = \sum_{i \leq m} \frac{1}{i} = \psi(m+1) - \psi(1)$ the harmonic numbers and $H_m^{\text{odd}} = \sum_{i \leq m} \frac{1}{2i-1} = \frac{1}{2}(\psi(m + \frac{1}{2}) - \psi(\frac{1}{2}))$ the odd harmonic numbers [1, 5.4.13, 5.4.15].

Proposition 6.2. For $k \geq 1$, with $t_k = t_k(h)$,

$$B_k = \frac{\ell_k}{2k+1} - \frac{2k^2+1}{3(2k+1)} \sum_{j=1}^k \frac{w_{k,j}}{2} \left[\psi\left(t_{k+1} - \frac{h}{2} - 2j\right) - \psi\left(t_k - \frac{h}{2} - 2j + 1\right) \right].$$

The bracket equals $H_{t_{k+1}-2j-1} - H_{t_k-2j}$ for $h = 0$ and $2(H_{t_{k+1}-2j-1}^{\text{odd}} - H_{t_k-2j}^{\text{odd}})$ for $h = 1$. In particular B_k is rational, $B_1(0) = \frac{163}{180}$, and $S_0(10) = B_1(0) + P_2(8) + P_2(9) + P_2(10) = \frac{25493}{25200}$.

Proof. Sum Lemma 6.1 over the block, using $\sum_{n=A}^B \frac{1}{n-\frac{h}{2}-2j+1} = \psi(B-\frac{h}{2}-2j+2) - \psi(A-\frac{h}{2}-2j+1)$. $\square$

Proposition 6.3 (integral representation). For $k \geq 1$ let

$$W_k(x) = \sum_{j=1}^k w_{k,j} x^{4(k-j)} = \frac{4k}{\pi} \int_0^1 \theta^{1/2} (1-\theta)^{-1/2} (1-\theta+\theta x^4)^{k-1} d\theta.$$

Then

$$B_k = \frac{\ell_k}{2k+1} - \frac{2k^2+1}{3(2k+1)} \int_0^1 x^{2t_k-h-4k+1} \frac{1-x^{2\ell_k}}{1-x^2} W_k(x) dx.$$

Proof. Insert $\frac{1/2}{y-2j+1} = \int_0^1 x^{2n-h-4j+1} dx$ (with $y = n - \frac{h}{2}$) into the block sum and sum the geometric series over the block. The integral formula for W_k is $2k$ times the probability generating function of $k-1-X \sim \text{BetaBinomial}(k-1, \frac{3}{2}, \frac{1}{2})$ at x^4 , written as a binomial mixture over a $\text{Beta}(\frac{3}{2}, \frac{1}{2})$ parameter, whose normalising constant is $B(\frac{3}{2}, \frac{1}{2}) = \frac{\pi}{2}$. $\square$

Remark 6.4. For fixed k the summand $P_k(n - \frac{h}{2})$ has no rational antidifference in n : at infinity it equals $\frac{1}{2k+1} - \frac{k(2k^2+1)}{3(2k+1)n} + O(n^{-2})$, while $G(n+1) - G(n)$ has no term in $1/n$ for any rational function G . This is why the digamma function enters Proposition 6.2; in particular the ordinary rational Gosper certificate [5] does not exist.

7 Asymptotics

Let $h \in \{0, 1\}$, fix k , and write $\eta = (-1)^{k+h}$,

$$D = \frac{4}{3}k^3, \quad a_\eta = \frac{5}{3}k + \frac{1-\eta}{2}, \quad \text{so that} \quad 2t_k(h) - h - 2 = D + a_\eta$$

by (8), and $\ell_k = 2k^2 + 2k + \frac{1+\eta}{2}$. Everything below depends on h and the parity of k only through η . For $n = t_k(h) + r$ with $0 \leq r < \ell_k$, equation (9) with $y = n - \frac{h}{2}$ becomes

$$P_k\left(n - \frac{h}{2}\right) = \frac{1}{2k+1} \left[1 - \left(1 + \frac{1}{2k^2}\right) \mathbb{E}\left[\frac{1}{1 + \xi_r/D}\right] \right], \quad \xi_r = a_\eta + 2r - 4X, \quad (10)$$

because $\frac{2k(2k^2+1)}{3D} = 1 + \frac{1}{2k^2}$. Put $T_m = \sum_{r=0}^{\ell_k-1} \mathbb{E}[\xi_r^m]$. The beta-binomial law has $\mathbb{E}X = \frac{k-1}{4}$ and $\text{Var}(4X) = k^2 - 1$ [7], so

$$T_0 = \ell_k, \quad T_1 = \ell_k \mu, \quad T_2 = \ell_k \left[\mu^2 + \frac{\ell_k^2 - 1}{3} + k^2 - 1 \right], \quad \mu = 2k^2 + \frac{8}{3}k + 1.$$

In general T_m is, for each η , a polynomial in k of degree at most $2m+2$ with rational coefficients. It follows from the factorial moments $\mathbb{E}[X(X-1)\cdots(X-i+1)] = (k-1)(k-2)\cdots(k-i)(\frac{1}{2})_i/(2)_i$ and the power sums $\sum_{r < \ell_k} r^j$.

Lemma 7.1. For $k \geq 8$, $0 \leq r < \ell_k$ and $0 \leq X \leq k-1$ we have $|\xi_r/D| \leq 4/k$.

Proof. We have $-\frac{7}{3}k + 4 \leq \xi_r \leq 4k^2 + \frac{17}{3}k$, and both $(4k^2 + \frac{17}{3}k)/D = \frac{3}{k} + \frac{17}{4k^2}$ and $\frac{7}{3}k/D = \frac{7}{4k^2}$ are at most $\frac{4}{k}$. $\square$

Proposition 7.2. Let $\eta = (-1)^{k+h}$.

(a) $B_k = \frac{3}{2} - \frac{3}{4k} + \frac{3\eta}{8k^2} + O(k^{-3})$.

(b) For every $J \geq 0$ there are rational numbers $b_{j,\pm}$, independent of h , with $B_k = \sum_{j=0}^J b_{j,\eta} k^{-j} + O_J(k^{-J-1})$. Table 2 lists them for $j \leq 8$.

(c) For $k \geq 8$ and $M \geq 1$,

$$\left| B_k - \frac{1}{2k+1} \left[\ell_k - \left(1 + \frac{1}{2k^2} \right) \sum_{m=0}^M (-1)^m \frac{T_m}{D^m} \right] \right| \leq \frac{3}{2} k \frac{(4/k)^{M+1}}{1 - 4/k}.$$

Proof. Sum (10) over r and use $\frac{1}{1+u} = \sum_{m=0}^M (-u)^m + \frac{(-u)^{M+1}}{1+u}$. By Lemma 7.1 the remainder is at most $\frac{\ell_k}{2k+1} (1 + \frac{1}{2k^2}) \frac{(4/k)^{M+1}}{1-4/k}$, and $\ell_k \leq 3k^2$ gives (c). For each η , $T_m/D^m = O(k^{2-m})$ is a Laurent polynomial in k , so the bracket in (c) equals $k F_{M,\eta}(1/k)$ for a polynomial $F_{M,\eta}$ with rational coefficients, and $B_k = F_{M,\eta}(1/k)/(2 + 1/k) + O(k^{-M})$. The term with index m contributes only from the power k^{1-m} on, so the Taylor coefficients of $F_{M,\eta}(z)/(2+z)$ up to z^{M-1} do not depend on M . This gives (b).

For (a) take $M = 3$. The formulas for T_1 and T_2 and $T_3 = \sum_r \mathbb{E}[\xi_r^3] = 32k^8 + O(k^7)$ give

$$\frac{\ell_k}{2k^2} = 1 + \frac{1}{k} + O(k^{-2}), \quad \frac{T_1}{D} = 3k + 7 + \frac{25 + 3\eta}{4k} + O(k^{-2}), \quad \frac{T_2}{D^2} = 6 + \frac{21}{k} + O(k^{-2}), \quad \frac{T_3}{D^3} = \frac{27}{2k} + O(k^{-2}).$$

The bracket in (c) equals $-\frac{\ell_k}{2k^2} + (1 + \frac{1}{2k^2}) (\frac{T_1}{D} - \frac{T_2}{D^2} + \frac{T_3}{D^3}) = 3k - \frac{3(1-\eta)}{4k} + O(k^{-2})$. Dividing by $2k+1$ gives (a). $\square$

j	$b_{j,+}$	$b_{j,-}$
0	3/2	3/2
1	-3/4	-3/4
2	3/8	-3/8
3	-9/20	57/40
4	291/160	-219/160
5	-2859/560	-327/280
6	6417/560	17793/2240
7	-203421/8960	-168561/8960
8	390009/8960	274299/8960

Table 2: Coefficients of the expansion $B_k \sim \sum_j b_{j,\eta} k^{-j}$ of Proposition 7.2, with $\eta = (-1)^{k+h}$: the column $b_{j,+}$ serves even k for $h = 0$ and odd k for $h = 1$. The certificates store the coefficients for $j \leq 59$.

Put $e_k = e_k(h) = B_k - \frac{3}{2} + \frac{3}{4k}$. By Proposition 7.2(a), $e_k = (-1)^{k+h} \frac{3}{8k^2} + O(k^{-3})$, so $\sum_k e_k$ converges absolutely, and $\sum_{k>K} e_k = O(K^{-2})$.

Theorem 7.3. Let $h \in \{0, 1\}$, $N \geq 3$, and let K and τ be as in Theorem B. Then

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O(K^{-2}), \quad (11)$$

uniformly in τ , where

$$\mathcal{A}(h) = \frac{3}{2} + \frac{3}{4}\gamma + \frac{1}{4} \log \frac{3}{2} - \sum_{k=1}^{\infty} \left(B_k(h) - \frac{3}{2} + \frac{3}{4k} \right). \quad (12)$$

At the block ends $N_K = t_{K+1}(h) - 1$ this reads $S_h(N_K) = \kappa N_K^{1/3} - \frac{1}{4} \log N_K - \mathcal{A}(h) - \frac{1}{4K} + O(K^{-2})$.

Proof. Block ends. By Theorem 5.4, $S_h(N_K) = \sum_{k \leq K} B_k = \frac{3}{2}K - \frac{3}{4}H_K + \sum_{k \leq K} e_k$, where $H_K = \log K + \gamma + \frac{1}{2K} + O(K^{-2})$. Hence

$$S_h(N_K) = \frac{3}{2}K - \frac{3}{4} \log K - \frac{3}{4}\gamma - \frac{3}{8K} + \sum_{k \geq 1} e_k + O(K^{-2}).$$

By (8), $N_K = \frac{2}{3}(K+1)^3 + \frac{5}{6}(K+1) + O(1)$, so $\kappa N_K^{1/3} = \frac{3}{2}(K+1) + \frac{5}{8K} + O(K^{-2})$ and $\log N_K = \log \frac{2}{3} + 3 \log K + \frac{3}{K} + O(K^{-2})$. Combining these expansions with (12) gives the block-end formula, because $-\frac{3}{8} - \frac{5}{8} + \frac{3}{4} = -\frac{1}{4}$.

General N . The case $K = 1$ is a finite statement, so let $K \geq 2$. Put $m = t_K(h) - 1 = N_{K-1}$, $r = N - m$ and $D_K = t_{K+1}(h) - t_K(h) = 2K^2 + 2K + O(1)$, so that $\tau = r/D_K$. The block-end formula for $K - 1$ gives $S_h(m) = \kappa m^{1/3} - \frac{1}{4} \log m - \mathcal{A}(h) - \frac{1}{4K} + O(K^{-2})$. For $1 \leq s \leq r$, equation (10) with $\xi = a_\eta + 2s - 2 - 4X$ applies to $n = t_K(h) - 1 + s$. Here $\mathbb{E}\xi = \frac{2}{3}K + 2s - \frac{1+\eta}{2}$ and $\mathbb{E}\xi^2 = (\mathbb{E}\xi)^2 + K^2 - 1$, and expanding $\frac{1}{1+u}$ to second order with Lemma 7.1 gives, uniformly in $1 \leq s < D_K$,

$$P_K\left(t_K(h) - 1 + s - \frac{h}{2}\right) = \frac{3s}{4K^4} - \frac{3s}{8K^5} - \frac{9s^2}{8K^7} + O(K^{-4}).$$

Summing over $s \leq r$ and inserting $r = \tau D_K$,

$$S_h(N) - S_h(m) = \frac{3}{2}\tau^2 + \frac{\frac{9}{4}\tau^2 - 3\tau^3}{K} + O(K^{-2}).$$

Since $r/m = O(1/K)$, Taylor expansion gives

$$\kappa N^{1/3} - \kappa m^{1/3} = \frac{3}{2}\tau + \frac{3(\tau - \tau^2)}{2K} + O(K^{-2}), \quad \frac{1}{4} \log \frac{N}{m} = \frac{3\tau}{4K} + O(K^{-2}).$$

Adding up, the coefficient of $1/K$ is $-\frac{1}{4} - \frac{3}{4}\tau + \frac{15}{4}\tau^2 - 3\tau^3 = -\frac{1}{4}(1 - 3\tau(1 - \tau)(4\tau - 1))$, and the terms of order one give $-\frac{3}{2}\tau(1 - \tau)$. $\square$

Theorem B follows, because $K \asymp N^{1/3}$.

8 The Alper constant

Definition 8.1. For $h \in \{0, 1\}$ let $N_K = t_{K+1}(h) - 1$ be the zeros of P , that is, the ends of the blocks. Put

$$\mathcal{A}(h) = \lim_{K \rightarrow \infty} \left(\kappa N_K^{1/3} - \frac{1}{4} \log N_K - S_h(N_K) \right).$$

The Alper constant is $\mathcal{A} = \mathcal{A}(0)$.

By Theorem 7.3 the limit exists and is given by the series (12). In the same way as Euler's constant measures how far the harmonic numbers lie from $\log N$, the Alper constant measures how far the equilibrium sums $S_0(N)$ lie from their smooth approximation $\kappa N^{1/3} - \frac{1}{4} \log N$; along general N the parabola $\frac{3}{2}\tau(1 - \tau)$ of Theorem B has to be added.

Representations

The series (12) has rational terms $e_k = B_k - \frac{3}{2} + \frac{3}{4k}$, which Proposition 6.2 evaluates exactly, and it converges absolutely. Its leading terms alternate and sum in closed form: since $\sum_k (-1)^{k+1} k^{-2} = \frac{\pi^2}{12}$, we have $\sum_k (-1)^{k+h} \frac{3}{8k^2} = (-1)^{h+1} \frac{\pi^2}{32}$. Thus

$$\sum_{k=1}^{\infty} e_k(h) = (-1)^{h+1} \frac{\pi^2}{32} + \sum_{k=1}^{\infty} \left(e_k(h) - (-1)^{k+h} \frac{3}{8k^2} \right),$$

and the last series, with terms $O(k^{-3})$, equals

$$\begin{aligned} &0.5655231753318100092599469738976772679775991822265765812903645817609304 \dots \quad (h = 0), \\ &0.0794352347285882502563253866959460981430671626443740113590515840927113 \dots \quad (h = 1). \end{aligned}$$

Subtracting further terms of Proposition 7.2 produces the contributions $\frac{381}{320}\zeta(3)$ and $-\frac{1497}{1280}\zeta(4)$ for $h = 0$, and $-\frac{69}{320}\zeta(3)$ and $\frac{2073}{1280}\zeta(4)$ for $h = 1$. Each such step is an exact identity and accelerates convergence, but we do not claim that the expansion in $1/k$ converges. Its coefficients in Table 2 grow quickly. Finally, Proposition 6.3 turns (12) into a series of explicit integrals.

Certified digits

Theorem 8.2. *The values of $\mathcal{A} = \mathcal{A}(0)$ and of $\mathcal{A}(1)$ printed in the introduction are correctly rounded to 80 decimal places. Moreover*

$$\begin{aligned} \sum_{k \geq 1} e_k(0) &\approx 0.2570980377977675524213691301515475449990 \\ &4607575030187421494741375405402095091354 \end{aligned}$$

is correctly rounded to 80 decimal places.

Method. For each h , split $\sum_k e_k$ at $K_0 = 192$. The 191 head terms come from Proposition 6.2. In the tail, B_k is replaced by the expansion of Proposition 7.2 through $j = 59$, with the exact rational coefficients computed from the polynomials T_m . The tail sums are Hurwitz zeta values: if a_0 and a_1 are the least even and odd integers $\geq K_0$, then $\sum_{k \geq K_0, k \equiv a_i(2)} k^{-j} = 2^{-j} \zeta(j, a_i/2)$. Two errors remain. Proposition 7.2(c) with $M = 60$ bounds the geometric truncation in each block. The Taylor truncation leaves $G(z)/(2+z)$, where $G = F_{60,\eta} - (2+z) \sum_{j \leq 59} b_{j,\eta} z^j$ is an explicit polynomial divisible by z^{60} , so it is at most $\frac{1}{2} \sum_p |g_p| k^{-p}$ in terms of the coefficients g_p of G . Both errors are summed over $k \geq K_0$ with $\sum_{k \geq K_0} k^{-s} \leq K_0^{-s} + K_0^{1-s}/(s-1)$. The two bounds are below $3.481 \cdot 10^{-100}$ and $1.26 \cdot 10^{-109}$ for both values of h .

All transcendental quantities (ψ , Hurwitz ζ , γ and $\log \frac{3}{2}$) are evaluated in Arb ball arithmetic [6] at 160 digits, via python-flint [4]. The resulting enclosures have radius below $7.4 \cdot 10^{-100}$, and the program checks that the rounding to 80 places is constant on each ball. For $h = 0$ a second run with $K_0 = 128$, the expansion through $j = 47$ and 110 digits gives an enclosure of radius below $6 \cdot 10^{-71}$ that overlaps the first. For both h , an independent 130-digit mpmath summation differs from the certified midpoint by less than 10^{-93} . This is a computer-assisted evaluation of the error bounds above, not a formal proof. $\square$

Is there a closed form?

We searched for an integer relation between $\mathcal{A}$ and

$$\begin{aligned} &1, \gamma, \log 2, \log 3, \pi, \pi^2, \pi^4, \zeta(3), \zeta(5), G, \\ &\log \pi, \log \Gamma(\tfrac{1}{4}), \log \Gamma(\tfrac{1}{3}), \sqrt{2}, \pi\sqrt{3}, \log^2 2, \log^2 3, \log 2 \log 3, \pi \log 2, \end{aligned}$$

where G is Catalan's constant, using PSLQ [3] in 120-digit arithmetic with tolerance 10^{-85} and coefficients of absolute value at most 1000. The same search was run for $\mathcal{A}(1)$, for $\mathcal{A}(h) - \frac{1}{4} \log \frac{3}{2}$ and for $\sum_k e_k(h)$, and also against the subset $1, \gamma, \log 2, \log 3, \pi^2, \zeta(3)$. No relation was found. These finite searches do not establish the absence of a closed form.

Two facts make $\mathcal{A}$ less arbitrary than an empirical constant. First, the block ends $t_k(0)$ are exact polynomials in k for each parity, by (8); the fractional positions of the zeros β_k never enter. Second, every ingredient of (12) is explicit: rational block sums and the constants γ and $\log \frac{3}{2}$.

How the constants were found

The constant $\mathcal{A}(1)$ was seen first, numerically. Exact block sums give $S_1(10^m)$ to many digits for $m \leq 20$. The defect $\kappa N^{1/3} - \frac{1}{4} \log N - S_1(N)$ oscillates inside each block. Subtracting $\frac{3}{2}\tau(1-\tau)$ from this defect leaves values clustering near 1.6464. The proofs did not use the special payments $2m-1$ except in the explicit form of the thresholds, which led to the family $G_n(h)$ and to its simplest member $h=0$, whose constant is the Alper constant. Theorem 7.3 and the certified evaluations turned these observations into theorems.

9 Numerical data

Table 3 lists $S_0(10^m)$ for $1 \leq m \leq 20$, computed in two ways. The certified computation uses Arb balls: for $m \leq 9$ it adds all block sums of Proposition 6.2, and for $m \geq 10$ it uses $S(N) = \frac{3}{2}(K-1) - \frac{3}{4}H_{K-1} + \sum_k e_k - \sum_{k \geq K} e_k$ plus the partial block, with the tail and the partial block taken from Proposition 7.2. It certifies 30 decimals. The second computation adds all block sums in 100-digit decimal arithmetic, switching to a bounded moment expansion of each block for $k \geq 1000$. The two agree to within $5 \cdot 10^{-31}$. The fourth column shows the normalised residual $K \delta(N)$ of the first-order formula, where

$$\delta(N) = S_0(N) - \left(\kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A} - \frac{3}{2}\tau(1-\tau) \right),$$

and the last column shows its limit $c(\tau) = -\frac{1}{4}(1 - 3\tau(1-\tau)(4\tau-1))$ predicted by Theorem 7.3. The difference between the two columns is $O(1/K)$, and it is below 10^{-5} for $m \geq 15$.

N	$S_0(N)$	$S_0(N)/N^{1/3}$	$K \delta(N)$	$c(\tau)$
10^1	1.01162698412698412698	0.469555651331	-0.164628	-0.268039
10^2	4.76382820302819364808	1.026335673796	-0.203027	-0.275258
10^3	13.30208678375677871897	1.330208678376	-0.089062	-0.151146
10^4	32.56774386666910234077	1.511660762675	0.041852	0.017932
10^5	74.87679399375861732781	1.613171624585	-0.287017	-0.291100
10^6	166.10210752872543471053	1.661021075287	-0.085058	-0.089064
10^7	363.77124561032651935744	1.688476551614	0.013286	0.010878
10^8	790.28053769454961122398	1.702607805266	-0.197141	-0.198360
10^9	1709.80689936028358017628	1.709806899360	0.034865	0.034247
10^{10}	3691.53391596882046613243	1.713458260326	-0.269046	-0.269115
10^{11}	7961.51938003709654146938	1.715257353771	-0.223351	-0.223461
10^{12}	17161.84551749443685308992	1.716184551749	-0.289401	-0.289424
10^{13}	36983.76129919255133262475	1.716634134713	-0.291166	-0.291168
10^{14}	79689.45681287532869805299	1.716857301875	-0.114766	-0.114791
10^{15}	171696.35811503857501272212	1.716963581150	-0.122306	-0.122312
10^{16}	369920.57702331276152428602	1.717019219635	-0.270994	-0.270996
10^{17}	796982.05832200729109138505	1.717045793782	-0.228871	-0.228872
10^{18}	1717058.94943545492359063880	1.717058949435	-0.254104	-0.254105
10^{19}	3699305.29382848385618983787	1.717065414396	-0.286261	-0.286261
10^{20}	7969925.78324778285942282425	1.717068458441	-0.016978	-0.016979

Table 3: $S_0(N)$ rounded to 20 decimals, the ratio $S_0(N)/N^{1/3}$, and the second-order residual of Theorem 7.3. The limit of the ratio is $\kappa = 1.7170713\dots$

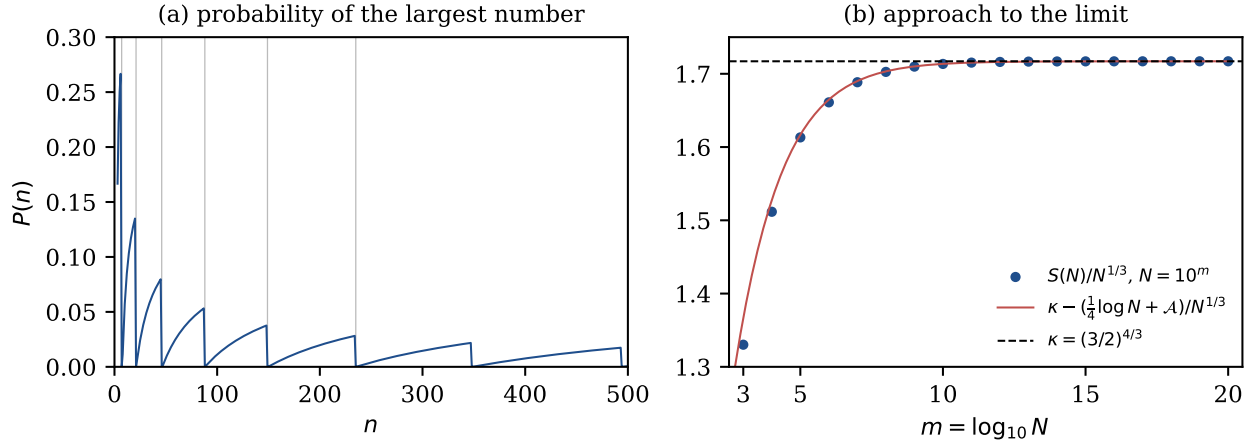


Figure 1: The game $G_n(0)$. (a) The probability $P(n)$ of the largest number for $3 \leq n \leq 500$; the grey lines mark the zeros $n = t_k(0) - 1$. Within the positive part of each block P increases. It drops to 0 one step before the support widens. (b) $S_0(N)/N^{1/3}$ at $N = 10^m$ for $3 \leq m \leq 20$ (dots), the limit κ (dashed), and the curve $\kappa - (\frac{1}{4} \log N + \mathcal{A})/N^{1/3}$.

10 Concluding remarks

The structure of the equilibria rests on two features of the payments: they grow linearly, which makes the payoff differences in Lemma 3.1 positive and the neighbour equations solvable, and the largest payment enters only through $y = n - \frac{h}{2}$. For $h \notin \mathcal{E} \cup \{0, 1\}$, Remark 5.5 proves that the thresholds eventually coincide with those of $h = 0$ or $h = 1$; the proof of Theorem 7.3 should then give the same

first-order formula, with a constant $\mathcal{A}(h)$ defined by the actual block sums, while the term of order $1/K$ acquires an h -dependent part. We have not pursued this.

We list some questions that we have not answered.

1. Does $\mathcal{A}$ have a closed form in classical constants? Is it irrational?
2. Can the asymptotic definition of $\mathcal{A}(h)$ be extended to every $h \notin \mathcal{E}$, and how does the resulting function depend on h ? The thresholds change at the points of $\mathcal{E}$, which accumulate at 1 and 2.
3. What is the nature of the series $\sum_j b_{j,\pm} k^{-j}$ in Table 2? Its coefficients grow quickly, and we expect it to diverge.
4. The method of Theorem 7.3 should give further terms of $S_h(N)$ in powers of $1/K$, with polynomial coefficients in τ . A general description of these polynomials would be interesting.

The exact checks of all displayed identities and examples, the computation of the tables and the figure, and the programs that certify $\mathcal{A}$ and $\mathcal{A}(1)$ are included in the source archive and at <https://github.com/AlperTheKing/alper-constant>. AI assistance was used in the derivations, in writing and running the verification code, and in preparing this manuscript. Review status: Author reviewed. The results have not been refereed or verified in a proof assistant. This is an unrefereed preprint.

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